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Manifolds for uniform vapor deposition

Abstract

A semiconductor device comprising a manifold for uniform vapor deposition is disclosed. The semiconductor device can include a manifold comprising a bore and having an inner wall. The inner wall can at least partially define the bore. A first axial portion of the bore can extend along a longitudinal axis of the manifold. A supply channel can provide fluid communication between a gas source and the bore. The supply channel can comprise a slit defining an at least partially annular gap through the inner wall of the manifold to deliver a gas from the gas source to the bore. The at least partially annular gap can be revolved about the longitudinal axis.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND OF THE INVENTION

Field

(2) The field relates generally to manifolds for uniform vapor deposition, and, in particular, to manifolds for improving reactant mixing in atomic layer deposition (ALD) reactors.

Description of the Related Art

(3) There are several vapor deposition methods for depositing thin films on surfaces of substrates. These methods include vacuum evaporation deposition, Molecular Beam Epitaxy (MBE), different variants of Chemical Vapor Deposition (CVD) (including low-pressure and organometallic CVD and plasma-enhanced CVD), and Atomic Layer Deposition (ALD).

(4) In an ALD process, one or more substrates with at least one surface to be coated are introduced into a deposition chamber. The substrate is heated to a desired temperature, typically above the condensation temperatures of the selected vapor phase reactants and below their thermal decomposition temperatures. One reactant is capable of reacting with the adsorbed species of a prior reactant to form a desired product on the substrate surface. Two, three or more reactants are provided to the substrate, typically in spatially and temporally separated pulses.

(5) In an example, in a first pulse, a first reactant representing a precursor material is adsorbed largely intact in a self-limiting process on a wafer. The process is self-limiting because the vapor phase precursor cannot react with or adsorb upon the adsorbed portion of the precursor. After any remaining first reactant is removed from the wafer or chamber, the adsorbed precursor material on the substrate reacted with a subsequent reactant pulse to form no more than a single molecular layer of the desired material. The subsequent reactant may, e.g., strip ligands from the adsorbed precursor material to make the surface reactive again, replace ligands and leave additional material for a compound, etc. In an unadulterated ALD process, less than a monolayer is formed per cycle on average due to steric hindrance, whereby the size of the precursor molecules prevent access to adsorption sites on the substrate, which may become available in subsequent cycles. Thicker films are produced through repeated growth cycles until the target thickness is achieved. Growth rate is often provided in terms of angstroms per cycle because in theory the growth depends solely on number of cycles, and has no dependence upon mass supplied or temperature, as long as each pulse is saturative and the temperature is within the ideal ALD temperature window for those reactants (no thermal decomposition and no condensation).

(6) Reactants and temperatures are typically selected to avoid both condensation and thermal

decomposition of the reactants during the process, such that chemical reaction is responsible for growth through multiple cycles. However, in certain variations on ALD processing, conditions can be selected to vary growth rates per cycle, possibly beyond one molecular monolayer per cycle, by utilizing hybrid CVD and ALD reaction mechanisms. Other variations may be allow some amount of spatial and/or temporal overlap between the reactants. In ALD and variations thereof, two, three, four or more reactants can be supplied in sequence in a single cycle, and the content of each cycle can be varied to tailor composition.

(7) During a typical ALD process, the reactant pulses, all of which are in vapor form, are pulsed sequentially into a reaction space (e.g., reaction chamber) with removal steps between reactant pulses to avoid direct interaction between reactants in the vapor phase. For example, inert gas pulses or “purge” pulses can be provided between the pulses of reactants. The inert gas purges the chamber of one reactant pulse before the next reactant pulse to avoid gas phase mixing. To obtain a self-limiting growth, a sufficient amount of each precursor is provided to saturate the substrate. As the growth rate in each cycle of a true ALD process is self-limiting, the rate of growth is proportional to the repetition rate of the reaction sequences rather than to the flux of reactant.

SUMMARY

(8) The systems and methods of the present invention have several features, no single one of which is solely responsible for its desirable attributes. Without limiting the scope of this invention as expressed by the claims which follow, various features will now be discussed briefly. After considering this discussion, and particularly after reading the section entitled “Detailed Description,” one will understand how the features described herein provide several advantages over traditional gas delivery methods and systems.

(9) In one embodiment, a semiconductor processing device is disclosed. The semiconductor processing device can include a manifold comprising a bore and having an inner wall, the inner wall at least partially defining the bore. A first axial portion of the bore can extend along a longitudinal axis of the manifold. The semiconductor processing device can include a supply channel that provides fluid communication between a gas source and the bore. The supply channel can comprise a slit defining an at least partially annular gap through the inner wall of the manifold to deliver a gas from the gas source to the bore. The at least partially annular gap can be revolved about the longitudinal axis.

(10) In another embodiment, a semiconductor processing device is disclosed. The semiconductor processing device can include a manifold comprising a bore and a supply channel that provides fluid communication between a gas source and the bore to supply a gas to the bore. The bore can comprise a channel having an annular flow portion with an at least partially annular cross-section and a non-annular flow portion with a non-annular cross-section, the non-annular flow portion disposed downstream of the annular flow portion.

(11) In another embodiment, a method of deposition is disclosed. The method can include supplying a gas through a supply channel to a bore of a manifold. The method can include creating an at least partially annular flow pattern in an annular flow portion of the bore such that the gas flows along a longitudinal axis of the manifold with an at least partially annular cross-section. Downstream of the annular flow portion, a non-annular flow pattern can be created in a non-annular portion of the bore such that the gas flows along the longitudinal axis with a non-annular cross-section.

(12) In another embodiment, a method of deposition is disclosed. The method can include supplying a gas to a supply channel. The method can include directing the gas from the supply channel to a bore of a manifold through a slit defining an at least partially annular gap along an inner wall of the manifold, the at least partially annular gap revolved about a longitudinal axis of the manifold.

(13) In another embodiment, a semiconductor processing device is disclosed. The semiconductor processing device can include a manifold comprising a bore therein, the bore defining a gas

passageway between a first end portion of the manifold and a second end portion of the manifold. The first end portion can be disposed opposite to and spaced from the second end portion along a longitudinal axis of the manifold by a first distance. The gas passageway can extend through the manifold for a second distance larger than the first distance. A reaction chamber can be disposed downstream of and in fluid communication with the bore.

(14) In another embodiment, a semiconductor processing device is disclosed. The semiconductor processing device can include a manifold comprising a bore having an axial portion that defines a longitudinal axis of the manifold and a lateral portion extending non-parallel to the longitudinal axis. The semiconductor processing device can include a supply channel that supplies gas to the axial portion of the bore at a first location along the longitudinal axis. The lateral portion can be disposed at a second location downstream of the first location, the lateral portion extending non-parallel relative to the longitudinal axis. The semiconductor processing device can include a reaction chamber disposed downstream of and in fluid communication with the bore.

(15) In another embodiment, a method of deposition is disclosed. The method can include providing a manifold comprising a bore therein. The bore can define a gas passageway between a first end portion of the manifold and a second end portion of the manifold. The first end portion can be disposed opposite to and spaced from the second end portion along a longitudinal axis of the manifold by a first distance. The method can comprise supplying a reactant gas to the bore. The method can comprise directing the reactant gas along the gas passageway from the first end portion to the second end portion for a second distance, the second distance larger than the first distance.

(16) In another embodiment, a method of deposition is disclosed. The method can include providing a manifold comprising a bore having an axial portion that defines a longitudinal axis of the manifold and a lateral portion extending non-parallel to the longitudinal axis. The method can include supplying a reactant gas to the axial portion of the bore at a first location along the longitudinal axis. The method can include directing the reactant gas through the axial portion of the bore parallel to the longitudinal axis. Downstream of the axial portion, the reactant gas can be directed through the lateral portion of the bore in a direction non-parallel to the longitudinal axis.

(17) In another embodiment, a semiconductor processing device is disclosed. The semiconductor processing device can include a manifold comprising a bore defining an inner wall a channel through the manifold and a source of gas. A supply channel can deliver the gas to the bore by way of an opening on the inner wall of the bore. All the gas can be delivered to the bore by the opening.

(18) In another embodiment, a method of deposition is disclosed. The method can include providing a manifold comprising a bore having an inner wall and defining a channel through the manifold. The method can include supplying all of a reactant gas through a single opening on the inner wall of the bore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects and advantages of the present invention will now be described with reference to the drawings of several embodiments, which embodiments are intended to illustrate and not to limit the invention.

(2) FIG. 1A is a schematic illustration of a flow path through a manifold of a semiconductor processing device.

(3) FIG. 1B is a schematic partial transverse cross-section of the flow path shown in FIG. 1A, taken along lines 1B-1B.

(4) FIG. 1C is a schematic top view of a gas deposition pattern on a substrate that is processed according to the flow path of FIGS. 1A-1B.

(5) FIG. 2 is a perspective view of an ALD manifold configured in accordance with various

embodiments.

(6) FIGS. 3A-3D illustrate an embodiment of a semiconductor device, in which a supply channel comprising a slit is used to supply gas to the bore.

(7) FIGS. 4A-4F illustrate another embodiment of a semiconductor device, in which the bore comprises a channel with an annular flow portion and a non-annular flow portion.

(8) FIGS. 5A and 5B are flowcharts illustrating example deposition methods, according to various embodiments.

(9) FIGS. 6A-6J illustrate an embodiment of a semiconductor processing device in which the manifold has an extended mixing length.

(10) FIGS. 7A and 7B are flowcharts illustrating example deposition methods, according to various embodiments.

(11) FIGS. 8A-8F illustrate various embodiments of a semiconductor processing device in which a single supply tier supplies gas to the bore.

(12) FIG. 9 is a flowchart illustrating an example deposition method, according to various embodiments.

DETAILED DESCRIPTION

(13) In vapor or gas deposition processes, it can be important to provide uniform deposition across the width or major surface of the substrate (e.g., a semiconductor wafer). Uniform deposition ensures that deposited layers have the same thickness and/or chemical composition across the substrate, which improves the yield of integrated devices (e.g., processors, memory devices, etc.), and therefore the profitability per substrate. To improve the uniformity of deposition, various embodiments disclosed herein can enhance the mixing profile of the different gases supplied within a manifold of the semiconductor processing system. Enhanced mixing of supplied gases can beneficially supply a relatively uniform gas mixture across the major surface of the substrate.

I. Overview of Atomic Layer Deposition Processes

(14) The embodiments disclosed herein can be utilized with semiconductor processing devices configured for any suitable gas or vapor deposition process. For example, the illustrated embodiments show various systems for depositing material on a substrate using atomic layer deposition (ALD) techniques. Among vapor deposition techniques, ALD has many advantages, including high conformality at low temperatures and fine control of composition during the process. ALD type processes are based on controlled, self-limiting surface reactions of precursor chemicals. Gas phase reactions are avoided by feeding the precursors alternately and sequentially into the reaction chamber. Vapor phase reactants are separated from each other in the reaction chamber, for example, by removing excess reactants and/or reactant by-products from the reaction chamber between reactant pulses. Removal can be accomplished by a variety of techniques, including purging and/or lowering pressure between pulses. Pulses can be sequential in a continuous flow, or the reactor can be isolated and can backfilled for each pulse.

(15) Briefly, a substrate is loaded into a reaction chamber and is heated to a suitable deposition temperature, generally at lowered pressure. Deposition temperatures are typically maintained below the precursor thermal decomposition temperature but at a high enough level to avoid condensation of reactants and to provide the activation energy for the desired surface reactions. Of course, the appropriate temperature window for any given ALD reaction will depend upon the surface termination and reactant species involved.

(16) A first reactant is conducted into the chamber in the form of vapor phase pulse and contacted with the surface of a substrate. Conditions are preferably selected such that no more than about one monolayer of the precursor is adsorbed on the substrate surface in a self-limiting manner. Excess first reactant and reaction byproducts, if any, are purged from the reaction chamber, often with a pulse of inert gas such as nitrogen or argon.

(17) Purging the reaction chamber means that vapor phase precursors and/or vapor phase byproducts are removed from the reaction chamber such as by evacuating the chamber with a

vacuum pump and/or by replacing the gas inside the reactor with an inert gas such as argon or nitrogen. Typical purging times for a single wafer reactor are from about 0.05 to 20 seconds, more preferably between about 1 and 10 seconds, and still more preferably between about 1 and 2 seconds. However, other purge times can be utilized if desired, such as when depositing layers over extremely high aspect ratio structures or other structures with complex surface morphology is needed, or when a high volume batch reactor is employed. The appropriate pulsing times can be readily determined by the skilled artisan based on the particular circumstances.

(18) A second gaseous reactant is pulsed into the chamber where it reacts with the first reactant bound to the surface. Excess second reactant and gaseous by-products of the surface reaction are purged out of the reaction chamber, preferably with the aid of an inert gas. The steps of pulsing and purging are repeated until a thin film of the desired thickness has been formed on the substrate, with each cycle leaving no more than a molecular monolayer. Some ALD processes can have more complex sequences with three or more precursor pulses alternated, where each precursor contributes elements to the growing film. Reactants can also be supplied in their own pulses or with precursor pulses to strip or getter adhered ligands and/or free by-product, rather than contribute elements to the film. Additionally, not all cycles need to be identical. For example, a binary film can be doped with a third element by infrequent addition of a third reactant pulse, e.g., every fifth cycle, in order to control stoichiometry of the film, and the frequency can change during the deposition in order to grade film composition. Moreover, while described as starting with an adsorbing reactant, some recipes may start with the other reactant or with a separate surface treatment, for example to ensure maximal reaction sites to initiate the ALD reactions (e.g., for certain recipes, a water pulse can provide hydroxyl groups on the substrate to enhance reactivity for certain ALD precursors).

(19) As mentioned above, each pulse or phase of each cycle is preferably self-limiting. An excess of reactant precursors is supplied in each phase to saturate the susceptible structure surfaces. Surface saturation ensures reactant occupation of all available reactive sites (subject, for example, to physical size or steric hindrance restraints) and thus ensures excellent step coverage over any topography on the substrate. In some arrangements, the degree of self-limiting behavior can be adjusted by, e.g., allowing some overlap of reactant pulses to trade off deposition speed (by allowing some CVD-type reactions) against conformality. Ideal ALD conditions with reactants well separated in time and space provide near perfect self-limiting behavior and thus maximum conformality, but steric hindrance results in less than one molecular layer per cycle. Limited CVD reactions mixed with the self-limiting ALD reactions can raise the deposition speed. While embodiments described herein are particularly advantageous for sequentially pulsed deposition techniques, like ALD and mixed-mode ALD/CVD, the manifold can also be employed for pulsed or continuous CVD processing.

(20) Examples of suitable reactors that may be used include commercially available ALD equipment such as any of the EmerALD® or Eagle® series reactors, available from ASM International of Almere, the Netherlands. Many other kinds of reactors capable of ALD growth of thin films, including CVD reactors equipped with appropriate equipment and means for pulsing the precursors, can be employed. In some embodiments a flow type ALD reactor is used, as compared to a backfilled reactor. In some embodiments, the manifold is upstream of an injector designed to distribute gas into the reaction space, particularly a dispersion mechanism such as a showerhead assembly above a single-wafer reaction space.

(21) The ALD processes can optionally be carried out in a reactor or reaction space connected to a cluster tool. In a cluster tool, because each reaction space is dedicated to one type of process, the temperature of the reaction space in each module can be kept constant, which improves the throughput compared to a reactor in which the substrate is heated to the process temperature before each run. A stand-alone reactor can be equipped with a load-lock. In that case, it is not necessary to cool down the reaction space between each run. These processes can also be carried

out in a reactor designed to process multiple substrates simultaneously, e.g., a mini-batch type showerhead reactor.

(22) FIG. 1A is a schematic illustration of a flow path **1** through a manifold of a semiconductor processing device. FIG. 1A illustrates the configuration of various channels inside the manifold, without showing the structure of the manifold itself, so as to better illustrate the relative orientation and interconnection of the internal channels of a manifold. The illustrated flow path **1** includes a bore **30** with an inert gas inlet **20** and an outlet **32**. The cross-sectional area of the bore **30** increases between the inlet **20** and the outlet **32**. In the illustrated arrangement, the cross-sectional area increases at a tapered portion **34**, which in the illustrated arrangement coincides with a merger of some of the reactant flow paths. The flow path **1** also includes a second inert gas inlet **22** which is in fluid communication with an inert gas distribution channel **40**. The inert gas distribution channel **40** extends generally in a plane intersecting the longitudinal axis of the bore **30**. Although the illustrated inert gas distribution channel **40** follows a circular curvature and extends a full 360°, in some embodiments, inert or reactant gas distribution channels can have other shapes (e.g., elliptical), and need not be a closed shape, that is, can extend only partway about the longitudinal axis of the bore, such as a C-shaped channel.

(23) The inert gas distribution channel **40** feeds inert gas to two inert gas passageways **42a**, **42b**, each of which can be connected to an inert gas valve. The inert gas passageways **42a**, **42b** connect with the inert gas distribution channel **40** at different angular locations distributed about the axis of the bore **30** (as viewed in a transverse cross-section). In the illustrated arrangement, the inert gas passageways **42a**, **42b** connect with the inert gas distribution channel **40** about 90° apart from one another, and about 135° (in opposite directions) from where the inert gas inlet **22** connects with the inert gas distribution channel **40**.

(24) The flow path **1** also includes a reactant gas passageway **37** which is in fluid communication with a reactant gas distribution channel **36**. The reactant gas distribution channel **36** extends generally in a plane intersecting the longitudinal axis of bore **30**, and is generally concentric with the inert gas distribution channel **40**. The reactant gas distribution channel **36** conveys gas to multiple, e.g., three reactant gas supply channels **38a**, **38b**, **38c** (only two of which are visible in FIG. 1A), each of which connects with the reactant gas distribution channel **36** at a different angular location about the axis of the bore **30** (as viewed in a transverse cross-section). In the illustrated embodiment, each of the reactant gas supply channels **38a**, **38b**, **38c** connect with the reactant gas distribution channel **36** at a location which is angularly offset from where the reactant gas passageway **37** connects with the reactant gas distribution channel **36**. The reactant gas supply channels **38a**, **38b**, **38c** also connect with the bore **30** at different angular locations distributed about the axis of the bore (as viewed in a transverse cross-section), and at an angle with respect to the longitudinal axis of the bore **30** (as viewed in a longitudinal cross-section).

(25) The flow path **1** also includes another reactant gas passageway **44b** which is in fluid communication with a reactant gas distribution channel **50**. The reactant gas distribution channel **50** extends generally in a plane intersecting the longitudinal axis of bore **30**. The reactant gas distribution channel **50** conveys reactant gas to multiple, e.g., three reactant gas supply channels **52a**, **52b**, **52c** (only two of which are visible in FIG. 1A), each of which connects with the reactant gas distribution channel **50** at a different angular location about the axis of the bore **30** (as viewed in a transverse cross-section). The reactant gas supply channels **52a**, **52b**, **52c** also connect with the bore **30** at different angular locations about the axis of the bore (as viewed in a transverse cross-section), and at an angle with respect to the longitudinal axis of the bore (as viewed in a longitudinal cross-section).

(26) The flow path **1** also includes a further reactant gas inlet **44a** which is in fluid communication with a reactant gas distribution channel **46**. The reactant gas distribution channel **46** extends generally in a plane intersecting the longitudinal axis of bore **30**. The reactant gas distribution channel **46** conveys reactant gas to multiple, e.g., three reactant gas supply channels **48a**, **48b**, **48c**,

each of which connects with the reactant gas distribution channel **46** at a different angular location about the axis of the bore **30** (as viewed in a transverse cross-section). The reactant gas supply channels **48a**, **48b**, **48c** also connect with the bore **30** at different angular locations about the axis of the bore **30** (as viewed in a transverse cross-section), and at an angle with respect to the longitudinal axis of the bore **30** (as viewed in a longitudinal cross-section). Each of the reactant gas supply channels **48a**, **48b**, **48c** connects with the bore **30** at a location which is angularly offset from where the reactant gas supply channels **52a**, **52b**, **52c** connect with the bore. The reactant gas supply channels **48a**, **48b**, **48c** also connect with the bore **30** at a greater angle than the reactant gas supply channels **52a**, **52b**, **52c** due to the reactant gas distribution channel **46** being a greater distance from the bore **30** than the reactant gas distribution channel **50**. Additionally, the bore **30** widens at the tapered portion **34** where the reactant gas supply channels **52a**, **52b**, **52c**, **48a**, **48b**, **48c** merge with the bore **30**. This allows a smoother merger and mixing of the reactants entering at this point with flow of gas (e.g., inert gas) that enters at upstream portions of the bore **30**.

(27) FIG. **1B** is a schematic partial transverse cross-section of the flow path shown in FIG. **1A**, taken along lines **1B-1B**. As shown in FIG. **1B**, the reactant gas supply channels **38a**, **38b**, **38c** connect with the bore **30** at different angular locations about the axis of the bore. As also shown in FIG. **1B**, horizontal components of the reactant gas supply channels **38a**, **38b**, **38c** extend in a radial direction from the axis (or, from the center) of the bore. The horizontal components of the reactant gas supply channels **52a**, **52b**, **52c** and the reactant gas supply channels **48a**, **48b**, **48c** can also connect with the bore in a radial fashion. Here, “horizontal” is meant to convey components of the supply channels in the plane of the cross-section, transverse to the bore axis, rather than any particular orientation relative to ground.

(28) Thus, in the flow pathway **1** shown in FIGS. **1A-1B**, a reactant gas pulse can deliver reactant gas through three separate supply channels and openings to the bore **30**. For example, in one pulse, a first reactant gas can be supplied to the bore **30** by way of the supply channels **38a**, **38b**, **38c**. In another pulse, a second reactant gas can be supplied to the bore **30** by way of the supply channels **52a**, **52b**, **52c**. In a third pulse, a third reactant gas can be supplied to the bore **30** by way of the supply channels **48a**, **48b**, **48c**. Additional details of the flow path **40**, and the semiconductor processing devices that define the flow path **40**, can be found throughout U.S. patent application Ser. No. 13/284,738, filed Oct. 28, 2011, the contents of which are incorporated by reference herein in their entirety and for all purposes.

(29) FIG. **1C** is a schematic top view of a gas deposition pattern **90** on a substrate that is processed according to the flow path **1** of FIGS. **1A-1B**. As shown in FIG. **1C**, the deposition pattern **90** includes three distinct spots **92** of regions with high concentrations of reactant gas mixtures, with the surrounding regions at lower concentrations. The three distinct spots **92** may result from the use of three distinct openings to the bore **30** that are in communication with three separate supply channels (such as supply channels **52a-52c**, **48a-48c**, **38a-38c**) that convey the same reactant gas to the bore **30** and ultimately to the substrate. Such non-uniform deposition may be undesirable, because different regions of the substrate may have different deposition chemistries and/or thicknesses, which can ultimately reduce device yield. Accordingly, there remains a continuing need for improving the uniformity of vapor deposition in semiconductor processing devices.

II. Manifolds with Annular Supply Slit and/or Annular Flow Pathways

(30) In some embodiments, vapor deposition uniformity can be improved by providing an at least partially annular slit in an inner wall of the bore to supply gases to the bore. For example, in various embodiments, the bore can comprise a first axial portion extending along a longitudinal axis of the manifold. A supply channel can be in fluid communication between a gas source (e.g., a reactant gas source) and the bore. The supply channel can comprise a slit defining an at least partially annular gap through the inner wall of the bore to deliver a gas from the gas source to the bore. The at least partially annular gap can be revolved about the longitudinal axis of the manifold.

(31) In addition, or alternatively, an at least partially annular flow pathway can be created in the

bore to deliver gases along a longitudinal axis of the manifold. For example, a supply channel can be in fluid communication between a gas source (e.g., a reactant gas source) and the bore. The bore can comprise a channel having an annular flow portion with an at least partially annular cross-section and a non-annular flow portion with a non-annular cross-section. The non-annular cross-section can be disposed downstream of the annular flow portion.

(32) FIG. 2 is a perspective view of an ALD manifold **100** configured in accordance with various embodiments. Unless otherwise noted, the components of FIG. 2 may be generally similar to the components of FIG. 1, except like components have been incremented by 100 relative to FIG. 1. As shown in FIG. 2, the manifold **100** comprises a body **102** that includes four blocks: an upper block **104**, an intermediate block **106**, a lower block **108** (see FIG. 3A), and a diffuser block **110**.

Although FIG. 2 shows a composite manifold body **102** comprising multiple stacked sub-portions or blocks, some embodiments can comprise fewer or more sub-portions or blocks, while others can comprise a monolithic or unitary manifold body. The use of multiple blocks **104**, **106**, **108**, **110** can beneficially enable the construction of channels disposed at various angles inside the manifold **100**.

(33) Mounted on the body **102** are two valve blocks **112a**, **112b**. An inert gas valve **114a** and a reactant gas valve **116a** are mounted on the valve block **112a**. An inert gas valve **114b** and a reactant gas valve **116b** are mounted on the valve block **112b**. Each of the valve blocks **112a**, **112b** can include a reactant gas inlet **118a**, **118b**. At upper block **104**, the manifold body **102** includes two inert gas inlets **120**, **122**. The reactant gas inlets **118a**, **118b** can be connected to different reactant sources, some of which may be naturally gaseous (i.e., gaseous at room temperature and atmospheric pressure), and some of which may be solid or liquid under standard conditions.

(34) The body **102** can also include one or more heaters **128**. Each of the valve blocks **112a**, **112b** can also include one or more heaters **126**. The heaters **126**, **128** can be disposed in such a manner as to maintain as constant a temperature as possible throughout the body **102** and/or the valve blocks. The heaters **126**, **128** can be any type of heater that can operate at high temperatures suitable for ALD processes, including without limitation linear rod-style, heater jacket, heater blank, heat trace tape, or coiled resistance heaters.

(35) FIG. 3A is a schematic side cross-sectional view of a semiconductor processing device **10** including the manifold **100** of FIG. 2, taken along lines 3A-3A of FIG. 2. As shown in FIG. 3A, the semiconductor processing device **10** can include the manifold **100** and a reaction chamber **810** disposed downstream of and coupled with the manifold body **102**. The manifold body **102** can comprise a longitudinal axis Z along which the bore **130** extends (or along which an axial portion of the bore **130** extends). In FIG. 3A, the inert gas inlet **120** at the top of the manifold body **102** connects with the bore **130** that extends longitudinally through the body **102** to an outlet **132**. The bore **130** has a larger cross-sectional area near the outlet **132** than it does near the inlet **120**. In the illustrated embodiment, the increase in cross-sectional area occurs at a tapered portion **134** of the bore **130**. Although not illustrated, an expander or other segment may be connected to the bottom of the manifold **100** to widen the flow path between the outlet **132** of the bore **130** and the reaction chamber **810**.

(36) A first reactant gas source **850a** can connect with a distribution channel **136** in the body **102** via a passageway **137**. The distribution channel **136** can be formed by lower and upper surfaces, respectively, of the upper block **104** and the intermediate block **106**, and can extend in a plane that intersects with the longitudinal axis of the bore **130**. For example, in some embodiments, the distribution channel **136** can be revolved at least partially (e.g., entirely) about the longitudinal axis Z of the manifold **100**. The distribution channel **136** can be in fluid communication with the bore **130** via a supply channel **138** comprising a slit through an inner wall **103** defined by the bore **130**. FIG. 3B illustrates examples of the slit formed through the inner wall **103**.

(37) The inert gas inlet **122** (see also FIG. 2) connects with an inert gas distribution channel **140** in the body **102**. The dashed line shown at the inlet **122** in FIG. 3A indicates that the passageway which connects the inlet **122** to the inert gas distribution channel **140** is not disposed in the cross-

section defined in FIG. 3A. An inert gas source **855** can supply an inert gas to the inert gas inlet **122** and the inert gas distribution channel **140**. The inert gas distribution channel **140** shown in FIG. 3A is formed by lower and upper surfaces, respectively, of the upper block **104** and the intermediate block **106**, and extends in a plane that intersects the longitudinal axis of the bore **130**. In some embodiments, the inert gas channel **140** can be disposed at about the same longitudinal location as the distribution channel **136**. The inert gas distribution channel **140** can supply inert gas to the inert gas valve **114a** via a passageway **142a**. The inert gas channel **140** may be revolved around the longitudinal axis Z, and may be disposed concentric relative to (e.g., concentrically about) the distribution channel **136**. As shown in FIGS. 2 and 3A, the passageway **142a** extends through the intermediate block **106** and the valve block **112a**. The inert gas distribution channel **140** can also supply inert gas to the inert gas valve **114b** via a passageway **142b**. The dashed lines for the passageway **142b** indicate that the passageway **142b** does not lie in the illustrated cross-section.

(38) With continued reference to FIG. 3A, the inert gas valve **114a** controls a supply of inert gas from the passageway **142a** (and thus, from the inert gas distribution channel **140**) to the reactant gas valve **116a**. The reactant gas valve **116a** controls a supply of a reactant gas from the inlet **118a** (or a mixture of a reactant gas from the inlet **118a** and an inert gas from the inert gas valve **114a**) to a passageway **144a**, which is connected to a gas distribution channel **146** in the body **102**. A second reactant source **850b** can supply a reactant gas to the inlet **118a**, the reactant gas valve **116a**, and the passageway **144a**. As shown in FIGS. 2 and 3A, the passageway **144a** extends through the valve block **112a**, the intermediate block **106**, and the lower block **108**. The distribution channel **146** can be formed by lower and upper surfaces, respectively, of the lower block **108** and the diffusion block **110**, and can extend in a plane that intersects the longitudinal axis Z of the manifold **100** (e.g., normal to the longitudinal axis Z in some embodiments). The distribution channel **146** can be in fluid communication with the bore **130** via a supply channel **148** comprising a slit (see FIG. 3B) through an inner wall **103** defined by the bore **130**.

(39) As shown in FIG. 3A, the inert gas valve **114b** can control a supply of inert gas from the passageway **142b** (and thus, from the inert gas distribution channel **140**) to the reactant gas valve **116b** (see FIG. 2). The reactant gas valve **116b** controls a supply of a reactant gas from the inlet **118b** (or a mixture of a reactant gas from the inlet **118b** and an inert gas from the inert gas valve **114b**) to a passageway **144b**, which is connected to a distribution channel **150** in the body **102**. The dashed lines in FIG. 3A indicate that the passageways **142b**, **144b** do not lie in the cross-section illustrated in FIG. 3A. A third reactant source **850c** can supply a reactant gas to the inlet **118b**, the reactant gas valve **116b**, and the passageway **144b**. As shown in FIG. 3A, the passageway **144b** extends through the valve block **112b** and the intermediate block **106**. The distribution channel **150** and/or the passageway **144b** can be formed by lower and upper surfaces, respectively, of the intermediate block **106** and the lower block **108**, and can extend in a plane that intersects the longitudinal axis Z of the manifold **100** (e.g., normal to the longitudinal axis Z in some embodiments). The distribution channel **150** can be in fluid communication with the bore **130** via a supply channel **152** comprising a slit (see FIG. 3B) through an inner wall **103** defined by the bore **130**. As shown in FIG. 3A, the distribution channel **150** and the supply channel **152** can be disposed and can connect with the bore **130** at a location along the longitudinal axis Z that is upstream of the distribution channel **146** and the supply channel **148**.

(40) While illustrated with three reactant inlets and two inert gas inlets to the manifold body **102**, the number of precursor/reactant and inert gas inlets can vary in embodiments. Also, while illustrated with two each, the number of precursor/reactant valves **116a**, **116b** and inert gas valves **114a**, **114b** feeding distribution channels can vary in embodiments, depending on the particular application and the desired processing capability of the ALD system. An ALD system may include at least two reactants and gas distribution channels therefor. The valves **114a**, **114b**, **116a**, and **116b** may be any type of valve that can withstand high temperatures within the ALD hot zone. Valves

114a, **114b**, **116a**, and **116b** may be ball valves, butterfly valves, check valves, gate valves, globe valves or the like. Metal diaphragm valves may also be used, and may be preferred for a high temperature environment (e.g., in temperatures up to about 220° C.). In some embodiments, the valves **114a**, **114b**, **116a**, and **116b** can be, for example and without limitation, pneumatically actuated valves or piezoelectric solenoid type valves. In embodiments, the valves **114a**, **114b**, **116a**, and **116b** can be configured to operate at very high speeds, for example, with opening and closing times of less than 80 ms, with speeds of less than 10 ms in some embodiments. The valves **114a**, **114b**, **116a**, and **116b** may be formed from any material that will function at the high temperatures required for ALD processing, such as 316L stainless steel and the like. Some embodiments, such as an ALD system configured for alumina deposition, can include valves configured to operate up to 220° C. Still other embodiments can include valves configured to operate at temperatures up to 300° C., up to 400° C., or at even higher temperatures.

(41) The manifold body **102** of FIG. 3A can be connected upstream of the reaction chamber **810**. In particular, the outlet **132** of the bore **130** can communicate with a reactant injector, particularly a dispersion mechanism in the form of a showerhead **820** in the illustrated embodiment. The showerhead **820** includes a showerhead plate **822** that defines a showerhead plenum **824** or chamber above the plate **822**. The showerhead **820** communicates vapors from the manifold **100** to a reaction space **826** below the showerhead **820**. The reaction chamber **810** includes a substrate support **828** configured to support a substrate **829** (e.g., a semiconductor wafer) in the reaction space **826**. The reaction chamber also includes an exhaust opening **830** connected a vacuum source. While shown with a single-wafer, showerhead type of reaction chamber, the skilled artisan will appreciate that manifold can also be connected to other types of reaction chambers with other types of injectors, e.g. batch or furnace type, horizontal or cross-flow reactor, etc.

(42) In the illustrated embodiment, three reactant sources **850a-850c** are shown, although fewer or greater numbers can be provided in other arrangements. In some embodiments, one or more of the reactant sources **850a-850c** can contain a naturally gaseous ALD reactant, such as H.sub.2, NH.sub.3, N.sub.2, O.sub.2, or O.sub.3. Additionally or alternatively, one or more of the reactant sources **850a-850c** can include a vaporizer for vaporizing a reactant which is solid or liquid at room temperature and atmospheric pressure. The vaporizer(s) can be, e.g., liquid bubblers or solid sublimation vessels. Examples of solid or liquid reactants that can be held and vaporized in a vaporizer include, without limitation, liquid organometallic precursors such as trimethylaluminum (TMA), TEMA₂Hf, or TEMA₂Zr; liquid semiconductor precursors, such as dichlorosilane (DCS), trichlorosilane (TCS), trisilane, organic silanes, or TiCl₄; and powdered precursors, such as ZrCl₄ or HfCl₄. The skilled artisan will appreciate that embodiments can include any desired combination and arrangement of naturally gaseous, solid or liquid reactant sources.

(43) As shown in FIG. 3A, the inert gas source **855** can provide purge gas to the reactant valves **116a**, **116b** and thus to the reactant distribution channels **146**, **150** (via the inert gas inlet **122**, distribution channel **140**, passageways **142a**, **142b** and inert gas valves **114a**, **114b**). The inert gas source **855** is shown feeding the top of the central bore **130** (via the inert gas inlet **120**). The same inert gas source **855** may also purge the reactant distribution channel **136** (via the reactant inlet **124** and the passageway **137**). However, in other embodiments, separate inert gas sources can be provided for each of these feeds.

(44) The semiconductor processing device **10** can also include at least one controller **860**, including processor(s) and memory with programming for controlling various components of the device **10**. While shown schematically as connected to the reaction chamber **810**, the skilled artisan will appreciate that the controller **860** communicates with various components of the reactor, such as vapor control valves, heating systems, gate valves, robot wafer carriers, etc., to carry out deposition processes. In operation, the controller **860** can arrange for a substrate **829** (such as a semiconductor wafer) to be loaded onto the substrate support **828**, and for the reaction chamber **810** to be closed, purged and typically pumped down in readiness for deposition processes, particularly atomic layer

deposition (ALD). A typical ALD sequence will now be described with reference to the reactor components of FIGS. 2 and 3A.

(45) In one embodiment, prior to reactant supply and during the entire ALD process, purge gas flows through the top inlet **120** into the bore **130**. When the controller instructs a first ALD reactant pulse, for example from the reactant source **850b**, the reactant valve **116a** is open to permit flow from the reactant source **850a** into the passageway **144a** and around the distribution channel **146**. Backpressure within the distribution channel **146** enables distribution of the gas through the supply channel **148** leading from the distribution channel **146** to the bore **130**, where the first reactant merges with the inert gas flow from the inlet **120**. At the same time, inert gas can flow through all other reactant channels (e.g., the reactant distribution channel **136**, the supply channel **138**, the reactant distribution channel **150** and supply channel **152**) into the bore **130**. From the bore **130**, the mixture of inert gas and first reactant is fed to the showerhead plenum **824** and distributed across the showerhead plate **822** (or other dispersion mechanism) and into the reaction space **826**. During this first reactant pulse, the narrower portion of the bore **130** upstream of the tapered portion **134** is filled with flowing inert gas and prevents upstream diffusion of the reactant.

(46) After a sufficient duration to saturate the substrate **829** surface with the first reactant, the controller **860** switches off the reactant valve **116a**, opens the inert gas valve **114a**, and thus purges the reactant valve **116a**, the passageway **144a**, the reactant distribution channel **146** and the depending supply channel **148**. Inert gas can continue to be supplied through the bore **130** from the top inlet **120** and the other reactant pathways for a sufficient duration to purge the manifold **100**, the showerhead plenum **824**, and the reaction space **826** of any remaining first reactant and/or byproduct. The skilled artisan will appreciate that other reactant removal procedures can be used in place of or in addition to purging.

(47) After a suitably long removal period to avoid interaction of the first reactant with the subsequent reactant, the controller **860** can instruct control valves to supply a second ALD reactant from, e.g., the gaseous reactant source **850a**, into the reactant passageway **137** and the upper reactant distribution channel **136**. Backpressure within the distribution channel **136** enables distribution of the gas through the supply channel **138** leading from the distribution channel **136** to the bore **130**, where the second reactant merges with the inert gas flow. At the same time, inert gas can flow through all other reactant channels (e.g., the reactant distribution channel **146**, the supply channel **148**, the reactant distribution channel **150** and supply channel **152**) into the bore **130**. From the bore **130**, the mixture of inert gas and second reactant is fed to the showerhead plenum **824** and distributed across the showerhead plate **822** (or other dispersion mechanism) and into the reaction space **826**. During this second reactant pulse, the portion of the bore **130** upstream of its merger with the supply channel **138** is filled with flowing inert gas, which prevents upstream diffusion of the second reactant. Similarly, the flow of inert gas through all other reactant flow paths prevents backwards diffusion of the second reactant.

(48) Following saturative surface reaction on the substrate, a removal step similar to the purge step described above can be performed, including purging of the distribution channel **136** and its depending supply channel **138**. The above described cycle can be repeated with the reactant distribution channel **150** and supply channel **152** to supply a third reactant gas to the substrate **829**. The cycle can be further repeated until a sufficiently thick layer is formed on the substrate **829**.

(49) FIG. 3B is a magnified side cross-sectional view of the portion 3B of the semiconductor processing device **10** shown in FIG. 3A. In particular, FIG. 3B shows the reactant distribution channels **146**, **150** which communicate with the bore **130** by way of the respective supply channels **148**, **152**. FIG. 3C is a schematic perspective cross-sectional view of a fluid pathway **101** defined through the distribution channels **146**, **150** and supply channels **148**, **152**. In particular, FIG. 3C is a negative of the portion 3B of the semiconductor device **10**, insofar as FIG. 3C illustrates the channels through which gas is supplied rather than the structure (e.g., the manifold **100**) which defines the channels.

(50) As shown in FIGS. 3B and 3C, the supply channels **148**, **152** can each comprise a respective slit **105a**, **105b** that defines an at least partially annular gap **107a**, **107b** in the inner wall **103** of the bore **130**. As shown in FIG. 3C, the slits **105a**, **105b** with associated gaps **107a**, **107b** can be revolved about the longitudinal axis Z of the manifold **100** such that the slits **105a**, **105b** define an at least partial annulus in the inner wall **103**. In the embodiment illustrated in FIGS. 3A-3C, the slits **105a**, **105b** extend entirely around the longitudinal axis Z, i.e., the slits **105a**, **105b** define a complete annulus of 360° revolution about the longitudinal axis Z. In other embodiments, however, the slits **105a**, **105b** can define a partial annulus about the longitudinal axis Z. For example, the slits **105a**, **105b** can be revolved about the Z axis by an angle in a range of 90° to 360°, in a range of 120° to 360°, in a range of 180° to 360°, in a range of 240° to 360°, etc.

(51) The gaps **107a**, **107b** defined by the slits **105a**, **105b** can comprise a narrow opening having a thickness less than a circumferential length of the slits **105a**, **105b**. That is, the arc length of the slits **105a**, **105b** along a perimeter or circumference of the inner wall **103** (i.e., about the axis Z) can be greater than the thickness of the gaps **107a**, **107b**. In some embodiments, the thickness of the gaps **107a**, **107b** can be in a range of 0.05 mm to 1.5 mm, or more particularly, in a range of 0.1 mm to 1 mm, in a range of 0.1 mm to 0.7 mm. In some embodiments, the thickness of the gaps **107a**, **107b** can be in a range of 0.05 mm to 0.5 mm, e.g., in a range of 0.1 mm to 0.5 mm, in a range of 0.1 mm to 0.3 mm, or in a range of 0.2 mm to 0.3 mm, or about 0.25 mm in some embodiments. In some embodiments, the thickness of the gaps **107a**, **107b** can be in a range of 0.3 mm to 1.5 mm, e.g., in a range of 0.3 mm to 1 mm, in a range of 0.3 mm to 0.7 mm, or in a range of 0.4 mm to 0.6 mm, or about 0.5 mm in some embodiments.

(52) By contrast, the thickness of the distribution channels **146**, **150** along the axis Z can be significantly greater than the thickness of the gaps **107a**, **107b**. For example, the thickness of the distribution channels **146**, **150** can be at least twice as thick as the gaps **107a**, **107b**, at least five times as thick as the gaps **107a**, **107b**, at least ten times as thick as the gaps **107a**, **107b**, at least twenty times as thick as the gaps **107a**, **107b**, or at least fifty times as thick as the gaps **107a**, **107b**. The gases inside the distribution channels **146**, **150** can have a backpressure caused by the restriction in thickness provided by the narrow gaps **107a**, **107b**. The backpressure can beneficially push the gases to the bore **130** through the gaps **107a**, **107b** of the slits **105a**, **105b**.

(53) FIG. 3D is a schematic side cross-sectional profile of the manifold body **102** and bore **130** illustrating the flow of a source gas S and an inert gas I during an example processing pulse. As shown in FIG. 3D, inert gas I (such as argon, Ar) can be supplied to the bore **130** by way of the central inert gas inlet **120** at the top of the manifold **100**. The inert gas I (e.g., Ar) can also be supplied to the bore **130** by the upstream supply channel **152**, which may also comprise a slit as explained above. In the illustrated arrangement, the inert gas I supplied through the bore **130** upstream of the supply channel **148** can comprise a push gas that drives gas through the bore **130** at sufficient pressure to cause the gases to rapidly travel to the reaction chamber **810**.

(54) As shown in FIG. 3D, the source gas S (e.g., titanium chloride) can be supplied through the slit **105a** of the supply channel **148** by way of the narrow gap **107a** in the inner wall **103** of the manifold **102** and defined by the bore **130**. The source gas S can be entrained with the inert gas I flowing downstream towards the reaction chamber **810**. As shown in FIG. 3D, because the source gas S is introduced along the periphery of the bore **130** through the slit **105a**, the source gas S may be concentrated about the outer edges of the central inert gas I flow path. Thus, as shown in FIG. 3D, the resulting downstream flow pattern can comprise a central inert gas I pattern disposed in the middle of the bore **130**, and an annular source gas S pattern disposed about the inert gas I pattern. The flow pattern of inert I and source S gases may comprise a non-annular pattern defined by the cross-section of the bore **130**. For example, the flow pattern of inert I and source S gases may mix together in a rounded (e.g., circular or elliptical) or polygonal cross-section defined by the cross-section of the bore **130**.

(55) The resulting patterns shown in FIG. 3D may be an improvement over the three-lobed flow

pattern shown in FIG. 1C. Instead of three different lobes representing high source gas concentration, in FIG. 3D, the concentration of the source and inert gases varies continuously without the three hot spots shown in FIG. 1C. For example, in FIG. 3D, the flow pattern is somewhat non-uniform and varies from the center of the bore **130** to the wall **103** of the bore **130**. However, adjustments to various processing temperatures may reduce the non-uniformity and enhance mixing.

(56) Although the supply channels with slits described in connection with FIGS. 3A-3D may improve uniformity of gas mixing in the manifold **100**, it may be desirable to further improve the mixing in order to improve device yield. FIGS. 4A-4F illustrate another embodiment of a semiconductor device **1**, in which the bore comprises a channel with an annular flow portion and a non-annular flow portion. Unless otherwise noted, the components of FIGS. 4A-4F may be similar to or the same as like numbered components in FIGS. 2-3D.

(57) In particular, FIG. 4A is a schematic side cross-sectional view of a semiconductor device **1** including the manifold **100** of FIG. 2, taken along the cross-section labeled 4A-4A. FIG. 4B is a schematic side cross-sectional view of the semiconductor device **1** of FIG. 4A, taken along the cross-section labeled 4B-4B in FIG. 2. The gas sources **850a-850c**, **855** and valves **114a-114b**, **116a-116b** are not shown in FIGS. 4A-4B for ease of illustration, but it should be appreciated that the channels in FIGS. 4A-4B can be connected to gas sources and valves in a manner similar to that shown in FIGS. 3A-3D.

(58) In FIGS. 4A and 4B, the manifold body **102** can comprise an inert gas distribution channel **140** which communicates with an inert gas source (such as source **855**) and with valves **114a-114b**, **116a-116b** by way of passageways **142a**, **142b** (see FIG. 3A). In addition, as with FIG. 3A, the manifold body **102** can comprise a first distribution channel **146** and a second distribution channel **150**. The first and second distribution channels **146**, **150** can be disposed at least partially about the longitudinal axis Z of the manifold **100**, and can be in fluid communication with corresponding reactant gas sources (such as the sources **850a-c**) and with the inert gas source, by way of passageways **144a**, **144b** and the reactant and inert gas valves. As explained above with respect to FIG. 3A, the inert gas valves **114a-114b** and the reactant gas valves **116a-116b** can be selectively activated to supply inert gas and/or reactant gas to the bore **130**. For example, as explained above with respect to FIGS. 3A-3D, the supply channels **148**, **152** can supply the gas to the bore **130** by way of corresponding slits **105a**, **105b** through the inner wall **103** of the bore **130**. The slits **105a**, **105b** can be revolved around the longitudinal axis Z to define an at least partial annulus (e.g., a complete annulus) in the wall **103** of the bore **130**.

(59) Unlike the embodiment of FIGS. 3A-3D, the lower block **108** can comprise three sub-blocks **108a**, **108b**, **108c**, in which the distribution channels **146**, **150** and supply channels **148**, **152** are formed. For example, as shown in FIG. 4A, the distribution channel **150** and supply channel **152** can be defined by a lower surface of the sub-block **108a** and an upper surface of the sub-block **108b**. The distribution channel **146** and supply channel **148** can be defined by a lower surface of the sub-block **108b** and an upper surface of the sub-block **108c**.

(60) In addition, as shown in FIGS. 4A-4B, a third distribution channel **171** can be defined in the diffusion block **110**, e.g., the channel **171** can be defined by a lower surface of the sub-block **108c** and an upper surface of the diffusion block **110**. The third distribution channel **171** can be revolved at least partially (e.g., entirely or partially) about the longitudinal axis Z and can be in fluid communication with a gas source (such as the reactant source **850a-850c** and/or the inert gas source **855**) by way of a passageway **175**. As with the distribution channels **146**, **150**, the distribution channel **171** can supply gas to the bore **130** by way of a supply channel **172**, which may be generally similar to the supply channels **148**, **152**. For example, the supply channel **172** can comprise a slit having an at least partially annular gap through the inner wall **103** of the bore **130**. The slit and gap can be revolved about the longitudinal axis Z as with the supply channels **148**, **152** described herein.

(61) Unlike the embodiment of FIGS. 3A-3D, the device **10** shown in FIGS. 4A-4B comprises an upstream non-annular flow pattern in an upstream non-annular flow portion **174A** of the bore **130**, an annular flow portion in an annular flow portion **173** of the bore **130**, and a downstream non-annular flow pattern in a downstream non-annular flow portion **174B** of the bore **130**. By contrast, the device **10** in FIGS. 3A-3D can comprise a non-annular flow portion through the length of the bore **130**. As shown in FIGS. 4A-4B, a plug **170** can be disposed within the bore **130** in the annular flow portion **173** of the bore **130**. As explained herein, the plug **170** can cooperate with the inner wall **103** of the manifold **100** to create an at least partially annular flow pathway as viewed in cross-section of the bore **130**.

(62) As used herein, the non-annular flow pattern and the non-annular flow portions **174A**, **174B** can comprise any suitable non-annular cross-section of the bore **130**. For example, the non-annular flow portions **174A**, **174B** can define a rounded (e.g., circular or elliptical) or a polygonal cross-section in which the gases fill the entire cross-section, e.g., there is no plug or obstruction in the non-annular flow portions **174A**, **174B**. Rather, the gases flow through the entire cross-section of the bore **130**.

(63) By contrast, the annular flow pattern and annular flow portion **173** can comprise an annular cross-section of the bore **130**, in which an interior region of the bore **130** is partially occluded so as to enable the gases to flow along the longitudinal axis **Z** through an annular region bounded by the inner wall **103** of the manifold **130** and an obstruction within the bore **130**, e.g., the plug **170**. The annular flow pattern and the cross-section of the annular portion **173** can be rounded (e.g., bounded by concentric circles or ellipses), polygonal (e.g., bounded by concentric polygons), or any other suitable annular shape. The annular cross-section may be symmetric in some embodiments. In other embodiments, the annular cross-section may be asymmetric.

(64) FIG. 4C is a schematic perspective view of the plug **170** coupled with the sub-block **108a** of lower block **108**. FIG. 4D is a schematic perspective cross-sectional view of the plug **170** and sub-block **108a** of FIG. 4C. Beneficially, the manifold body **102** can be formed of multiple blocks and sub-blocks, as explained herein. The modularity with which the manifold body **102** can be constructed enables the introduction of useful components such as the plug **170** shown in FIGS. 4C and 4D. For example, as shown in FIGS. 4C and 4D, the sub-block **108a** can include an opening **176** through which the plug **170** is disposed. In some embodiments, the plug **170** can be connected to the sub-block **108a** by way of an adhesive or fastener. In some embodiments, the plug **170** can be fitted into the opening **176** by way of an interference or friction fit. In still other embodiments, the plug **170** can comprise a nail-like feature, in which an upstream flange extends over the upper surface of the sub-block **108a** to secure the plug **170** to the sub-block **108a**.

(65) Moreover, as shown in FIGS. 4C-4D, the sub-block **108a** can include a plurality of holes **177** disposed about the opening **176** and the plug **170**. Thus, gases supplied upstream of the sub-block **108a** can pass around an upstream tapered portion **170A** of the plug **170**. The upstream tapered portion **170A** of the plug **170** can create a transition from non-annular flow to annular flow. As the gases approach the upstream tip of the tapered portion **170A**, the gases can be divided into an at least partially annular flow pattern. The holes **177** can enable upstream gases to pass through the sub-block **108a** and around the plug **170**. A downstream tapered portion **170B** of the plug **170** can transition the gases from annular flow to non-annular flow.

(66) FIG. 4E is a schematic perspective cross-sectional view of a flow pathway **178** defined through the distribution channels **146**, **150**, **171** and supply channels **148**, **152**, **172**. In particular, FIG. 4E is a negative of a portion of the manifold body **102**, insofar as the channels through which gas is supplied is illustrated rather than the structure (e.g., the manifold **100**) which defines the channels. As shown in FIG. 4E, at an upstream portion of the flow pathway **178**, the gas can flow along an at least partial annular pathway **173** through hole channels **177A** defined by the holes **177** of FIGS. 4C-4D. As explained above, the holes **177** can enable the gases to flow through the sub-block **108a**. Gas from the distribution channel **150** can be supplied to the annular portion **173** of the

bore **130** by way of the supply channel **152**, which can comprise the slit **105b**. Gas from the distribution channel **146** can be supplied to the annular portion **173** of the bore **130** by way of the supply channel **148** comprising the slit **105a**. Similarly, gas from the distribution channel **171** can be supplied to the annular portion **173** of the bore **130** by way of the supply channel **172** which can comprise a slit **105c**.

(67) The at least partial annular portion **173** shown in FIG. **4E** is a complete annulus (i.e., revolved about the longitudinal axis **Z** by 360° , but in other embodiments, the portion **173** can comprise a partial annulus defining a revolution between 90° and 360° , between 120° and 360° , between 180° and 360° , between 240° and 360° , etc. As shown in FIG. **4E**, the annular portion **173** can transition into the downstream non-annular pathway **174B**, and the gases can be conveyed to the reaction chamber **510** as explained herein.

(68) FIG. **4F** is a magnified, schematic side cross-sectional view of the manifold body **102**, illustrating the upstream non-annular flow portion **174A**, the annular flow portion **173**, and the downstream non-annular flow portion **174B**. As shown in FIG. **4F**, a first inert gas I.sub.1 (such as Argon) can be supplied through the inlet **120**. The first inert gas I.sub.1 can flow along the longitudinal axis **Z** through the non-annular flow portion **174A** of the bore with a non-annular flow cross-section. As shown on the right side of FIG. **4F**, the gases flowing through the non-annular portion **174A** can have a non-annular cross-sectional flow profile N.sub.1, in which the first inert gas I.sub.1 flows through the entire volume bounded by the inner wall **103** of the bore **130**. Thus, the flow profile N.sub.1 does not have any barriers or plugs in the interior of the bore **130**.

(69) The first inert gas I.sub.1 can transition from non-annular flow to at least partially annular flow (e.g., complete annular flow) within the annular flow portion **173** when the first inert gas I.sub.1 encounters the downstream tapered portion **170A** of the plug **170**. The first inert gas I.sub.1 can pass through the holes **177** and can travel downstream along the annular flow portion **173** about the outer periphery of the plug **170**, e.g., between the outer periphery of the plug **170** and the inner wall **103** of the manifold body **102**. As shown in a first annular flow profile A.sub.1, the first inert gas I.sub.1 can uniformly fill the annular space provided between the plug **170** and the inner wall **103** of the manifold body **102**.

(70) During an example pulse of gas to the device **10**, a source gas **S** can be supplied to the annular portion **173** of the bore **130** by way of the distribution channel **146** and the supply channel **148**. For example, as explained above, the source gas **S** (e.g., a reactant gas) can be delivered from the wider distribution channel **146** to the narrow slit **105a** by way of backpressure built up in the channel **146**. As shown in a second annular flow profile A.sub.2, the source gas **S** can enter uniformly about the wall **103** such that source gas **S** can push the inert gas I.sub.1 radially inward. In the second annular flow profile A.sub.2, the source gas **S** can be disposed concentrically about the inert gas I.sub.1. Beneficially, the annular flow portion **173** can promote mixing between the source gas **S** and the first inert gas I.sub.1, at least in part because the constricted area provided by the flow portion **173** causes the source gas **S** and the first inert gas I.sub.1 to intermix together.

(71) A second inert gas I.sub.2 (such as argon) can be supplied to the annular flow portion **173** of the bore **130** by way of the third distribution channel **171** and the supply channel **172**, which can comprise a narrow slit **105c** defining an at least partially annular gap **107c** through the wall **103** of the bore **130**. Advantageously, the second inert gas I.sub.2 can push the source gas **S** and the first inert gas I.sub.1 towards the outer periphery of the plug **170** to enhance mixing. As shown in FIG. **4F**, a third annular flow profile A.sub.3 can be defined in the annular portion **173** in which the source gas **S** is disposed annularly between the first and second inert gases I.sub.1, I.sub.2. The constricted annular portion **173** can enhance mixing among the source gas **S** and the inert gases I.sub.1, I.sub.2, as shown in a fourth annular flow profile A.sub.4 disposed about the downstream tapered portion **170B** of the plug **170**.

(72) The mixed gases can transition from an annular flow profile A.sub.4 to a second non-annular profile N.sub.2 downstream of the plug **170**. As the mixed gases emerge into the downstream non-

annular portion **174B**, the gases can be sufficiently mixed so as to provide a substantially uniform concentration and/or thickness on the substrate. Thus, the embodiment shown in FIGS. **4A-4F** can improve the mixing and reduce non-uniformities associated with other types of flow manifolds. The restricted flow paths defined by the annular portion **173** of the bore **130** can facilitate improved mixing of any number and type of gases. The embodiment of FIGS. **3A-3D** and **4A-4F** can result in an average deposition non-uniformity of less than 5%, e.g., less than 2% (e.g., about 1.8%), as compared with the pattern of FIG. **1C** which can result in an average non-uniformity of about 14%.

(73) FIG. **5A** is a flowchart illustrating a method **500** of depositing a film on a substrate. The method **500** begins in a block **501** to supply a reactant gas through a supply channel to a bore of a manifold. As explained herein, a distribution channel can convey gas from a source (such as a source of reactant or inert gas) to the supply channel. The distribution channel can be annularly disposed about a longitudinal axis of the manifold.

(74) In a block **502**, the reactant gas can be directed from the supply channel to the bore through a slit. The slit can define an at least partially annular gap along an inner wall of the bore. The at least partially annular gap can be revolved around the longitudinal axis. For example, the slit can comprise a full annulus revolved 360° about the longitudinal axis. In other embodiments, the slit can comprise a partial annulus revolved only partially about the longitudinal axis. As explained herein, the at least partially annular gap can comprise a thickness that is significantly smaller than a circumferential or peripheral length of the gap along the wall of the manifold. Beneficially, as explained herein, the slit can provide relatively uniform gas flow to the bore. In some embodiments, as explained herein, a plug can be provided to define an at least partially annular flow path. Non-annular flow paths can be provided upstream and downstream of the at least partially annular flow path.

(75) FIG. **5B** is a flowchart illustrating a method **550** of depositing a film on a substrate, according to various embodiments. The method **550** begins in a block **551** to supply a reactant gas through a supply channel to a bore of a manifold. As explained herein, a distribution channel can convey gas from a source (such as a source of reactant or inert gas) to the supply channel. The distribution channel can be annularly disposed about a longitudinal axis of the manifold in some embodiments.

(76) Moving to a block **552**, an at least partially annular flow pattern can be created in an annular flow portion of the bore such that the reactant gas flows along a longitudinal axis of the manifold with an at least partially annular cross-section. For example, in some embodiments, the at least partially annular flow pattern can be defined by a plug (such as the plug **170**) disposed within the bore. The plug can partially obstruct the bore to divide the gas flow such that the gas flows about an outer periphery of the plug. As explained herein, upstream of the at least partially annular cross-section, the gas can flow in an upstream non-annular flow pattern. When the gas reaches the annular flow portion, the gas can flow around the outer periphery of the plug. The constricted area provided by the annular flow pathway can beneficially enhance the mixing of gases flowing through the bore.

(77) In a block **553**, downstream of the annular flow portion, a non-annular flow portion can be created in a non-annular portion of the bore such that the reactant gas flows along the longitudinal axis with a non-annular cross-section. As explained herein, the plug can comprise upstream and downstream tapered portions that can enable the transition of the gas flow from non-annular to annular, and from annular to non-annular. The convergence of the annular gas pathway into a downstream non-annular portion can further enhance mixing of supplied gases, which can advantageously improve device yield.

III. Manifolds with Extended Mixing Length

(78) Various embodiments disclosed herein can enable reduce deposition non-uniformity and improve mixing by extending the mixing length along the bore **130** downstream of the location(s) at which gases are supplied to the bore **130**. For example, in some embodiments, a semiconductor processing device can comprise a manifold comprising a bore therein. The bore can define a gas

passageway between a first end portion of the manifold and a second end portion of the manifold. The first end portion can be disposed opposite to and spaced from the second end portion along a longitudinal axis of the manifold by a first distance. The gas passageway can extend through the manifold for a second distance larger than the first distance. For example, in some embodiments, the second distance can be at least 1.5 times the first distance, at least 2 times the first distance, at least 3 times the first distance, or at least 5 times the first distance. In some embodiments, the second distance can be in a range of 1.5 times to 10 times the first distance, e.g., in a range of 2 times to 5 times the first distance. A reaction chamber can be disposed downstream of and in fluid communication with the bore.

(79) In some embodiments, a semiconductor processing device can include a manifold comprising a bore having an axial portion that defines a longitudinal axis of the manifold and a lateral portion extending non-parallel to the longitudinal axis. A supply channel that supplies gas to the axial portion of the bore can be disposed at a first location along the longitudinal axis. The lateral portion can be disposed at a second location downstream of the first location. The lateral portion can extend non-parallel relative to the longitudinal axis. A reaction chamber can be disposed downstream of and in fluid communication with the bore.

(80) FIGS. 6A-6J illustrate an embodiment of a semiconductor processing device **10** in which the manifold **100** has an extended mixing length. Unless otherwise noted, reference numerals in FIGS. 6A-6J refer to components that are the same as or similar to like numbered components from FIGS. 2-4F. For example, FIG. 6A is a schematic perspective view of the manifold **100** with an extended mixing length. FIG. 6B is a schematic perspective exploded view of the manifold **100** of FIG. 6A. The manifold **100** can include a manifold body **102** connected with valve blocks **112a**, **112b**. Reactant valves **116a**, **116b** and inert gas valves **114a**, **114b** can be disposed on the blocks **112a**, **112b**. Inert gas inlets **120**, **122** can supply inert gas to the manifold **100**. The manifold body **102** can comprise multiple blocks **104**, **106**, **108**. Unlike the embodiment of FIGS. 3A-4F, the intermediate block **106** can comprise a sub-block **106a** and a sub-block **106b**. The lower block **108** can comprise a first sub-block **108a**, a second sub-block **108b**, and a third sub-block **108c**. As explained above, the use of multiple blocks and sub-blocks can enable a modular construction of the manifold **100** which can enable the use of internal channels with curved shapes and other internal lumens.

(81) Beneficially, as explained herein, the sub-blocks **108a-108c** can define an extending mixing length pathway **180** having a first lateral portion **180a**, an offset axial portion **180b**, and a second lateral portion **180b**. As explained herein, the pathway **180** can provide an extended mixing length downstream of where the supply gases are introduced to the bore **130**.

(82) FIG. 6C is a schematic side cross-sectional view of a semiconductor processing device **10** that includes the manifold **100** of FIGS. 6A-6B and a reaction chamber **810**. As with FIGS. 1A and 2-4F, the manifold **100** can include gas distribution channels **136**, **150**, and **146**, in addition to an inert gas distribution channel **140**. Supply channels **138a-c** can convey gases from the distribution channel **136** to the bore **130**. Supply channels **152a-c** can convey gases from the distribution channel **150** to the bore **130**. Supply channels **148a-c** can convey gases from the distribution channel **146** to the bore **130**. In the embodiment of FIG. 6C, the supply channels **138a-c**, **152a-c**, **148a-c** can comprise angled supply channels similar to those shown in FIG. 1A. In other embodiments, however, the supply channels can comprise the supply channels **138**, **148**, **152**, and/or **172** of FIGS. 3A-4F, which include slits defining an at least partially annular gap about the longitudinal axis Z of the manifold **100**. Moreover, in some embodiments, the bore **130** of FIG. 6C can comprise a plug **170** defining an at least partially annular flow pathway through which the gases can flow. Thus, the manifold **100** of FIG. 6C can be used in combination with the slits and/or annular flow portions described in connection with FIGS. 3A-4F.

(83) As explained below in connection with FIGS. 6D-6J, the extended mixing length pathway **180** can extend the mixing length of the bore **130** to enhance mixing. As shown in FIG. 6C, the pathway

180 of the bore **130** can be disposed downstream of the location **L** at which the most downstream supply channel supplies gas to the bore **130**. Thus, as explained herein, the gases supplied by the supply channels **138**, **146**, **152** may initially mix within an upstream axial portion **130A** of the bore **130** which extends along the longitudinal axis **Z** upstream of the extended mixing length pathway **180**.

(84) The pathway **180** can extend the mixing length (and therefore mixing time) of the supplied gases as compared with a bore that extends straight through the manifold **100** along the longitudinal axis **Z**. As explained herein, the extended length pathway **180** can comprise the first lateral portion **180a** which extends non-parallel to and away from the longitudinal axis **Z**, the offset axial portion **180b** which extends generally parallel to, but offset from, the longitudinal axis **Z**, and the second lateral portion **180c** which extends non-parallel to and towards the longitudinal axis **Z**. The second lateral portion **180c** of the bore **130** can transition into a downstream axial portion **130B** that extends downstream from the pathway **180** along the longitudinal axis **Z** to the reaction chamber **810**. Although the downstream axial portion **130B** is illustrated as being disposed within the manifold **100** for some length, it should be appreciated that the downstream axial portion **130B** may comprise a very short length or may comprise a juncture at which the pathway **180** merges with the bore **130** at the inlet to the reaction chamber **810**. That is, the second lateral portion **180c** may extend laterally towards the axis **Z**, and an opening in the manifold may provide axial fluid communication directly between the second lateral portion **180c** and the reaction chamber **810**. In such an embodiment, the downstream axial portion **130B** can comprise the opening or aperture which provides axial fluid communication between the pathway **180** and the reaction chamber **810**.

(85) FIGS. **6D-6J** illustrate the extended mixing length flow pathway **180** which is disposed downstream of the location **L** at which the downstream-most supply channel merges with the bore **130**. In particular, FIGS. **6A-6I** illustrate the pathway **180** through the sub-blocks **108a-108c**. FIG. **6J** is a schematic perspective view of the flow pathways through the manifold **100**.

(86) For example, FIG. **6D** is a schematic top perspective view of the first sub-block **108a**. FIG. **6E** is a schematic bottom perspective view of the first sub-block **108a**. As shown in FIGS. **6D** and **6J**, the bore **130** can comprise an upstream axial flow portion **130A** disposed upstream of and terminating in the sub-block **108a**. The upstream axial flow portion **130A** can extend along the longitudinal axis **Z** of the manifold **100**. Although the longitudinal axis **Z** of the manifold **100** is illustrated as being disposed perpendicular to the top surface of the manifold **100**, in other embodiments, the longitudinal axis **Z** can be disposed obliquely through the manifold **100**. The upstream axial flow portion **130A** of the bore **130** can be disposed generally parallel to or along the axis **Z**. As shown in FIG. **6D**, the upstream axial flow portion **130A** enters an upper surface **181** of the first sub-block **108a**.

(87) In FIGS. **6E** and **6J**, the upstream axial flow portion **130A** can extend axially (i.e., along the longitudinal axis **Z**) through a portion of a thickness of the sub-block **108a**. FIG. **6F** is a top schematic perspective view of the second sub-block **108b**. When assembled (see FIG. **6C**), the first sub-block **108a** and the second sub-block **108b** can connect together by way of one or more mechanical fasteners. The first and second sub-blocks **108a**, **108b** can cooperate to define the first lateral portion **180a** of the pathway **180** of the bore **130**. For example, a lower surface **182** of the first sub-block **108a** can comprise a first groove **183** formed through a portion of the thickness of the first sub-block **108a**. The upper surface **181** of the second sub-block **108b** can comprise a second groove **184** formed through a portion of the thickness of the second sub-block **108b**. As shown in FIGS. **6E-6F**, the grooves **183**, **184** can extend from the axial pathway **130A** away from and non-parallel to the longitudinal axis **Z**. In the illustrated embodiment, the grooves **183**, **184** extend perpendicular to the axis **Z**. As shown in FIGS. **6E-6F**, the grooves **183**, **184** delimit a spiral pattern, beginning at the axial flow portion **130A** and curving outwardly to an outer portion of the sub-blocks **108a-108b**.

(88) FIG. **6G** is a schematic bottom perspective view of the second sub-block **108b**. As shown in

FIGS. 6C, 6F, 6G, and 6J, the offset axial portion **180b** can be disposed offset from the axis Z and can include a component generally parallel to the axis Z. In the illustrated embodiment, the offset axial portion **180b** is disposed generally parallel to the axis Z. In other embodiments, the offset axial portion **180b** may not be parallel to the axis Z, but rather may include a directional component parallel to the axis Z, such that the offset axial portion **180b** conveys the gas through at least some displacement along the axis Z. As shown in FIGS. 6F and 6G, the offset axial portion **180b** can be defined along an axial channel **185** formed through the entire thickness of the sub-block **108b**.

(89) FIG. 6H is a schematic top perspective view of the third sub-block **108c**. When assembled, the lower surface **182** of the second sub-block **108b** can cooperate with the upper surface **181** of the third sub-block **108c** to define the second lateral portion **180c** of the flow pathway **180**, as shown in FIG. 6J. For example, a third groove **186** can be formed in the lower surface **182** of the second sub-block **108b**. A fourth groove **187** can be formed in the upper surface **181** of the third sub-block **108c**. As shown in FIGS. 6G, 6H, and 6J, the grooves **186**, **187** and the second lateral portion **180c** can extend non-parallel to and towards the longitudinal axis Z in a spiral pattern. For example, the second lateral flow portion **180c** can extend laterally (e.g., non-parallel to the axis Z) from the offset axial channel **185** to a central portion of the sub-block **108c**.

(90) FIG. 6I is a schematic bottom perspective view of the third sub-block **108c**. As illustrated in FIGS. 6H-6J, the groove **187** of the third sub-block **108c** can communicate with the downstream axial flow portion **130B** of the bore **130**. Gases flowing through the second lateral portion **180c** can transition from lateral flow through the pathway **180** to axial flow through the axial portion **130B** of the bore **130**.

(91) In the illustrated embodiments, the extended length flow pathway **180** extends laterally away from the longitudinal axis Z, extends parallel to but offset from the axis Z, and extends laterally towards the longitudinal axis Z. In the illustrated embodiments, the first and second axial portions **130A**, **130B** of the bore **130** are generally aligned along the longitudinal axis Z. However, it should be appreciated that in other embodiments, the downstream axial portion **130B** may be offset from the longitudinal axis Z. For example, in such embodiments, the reaction chamber **810** and the outlet **132** may be disposed offset from the inlet **120** and the axis Z. Moreover, in the illustrated embodiments, the pathway **180** includes two lateral portions **180a**, **180c**, and one offset axial portion **180b**. In other embodiments, however, additional sub-blocks may be added to provide additional mixing length. For example, in such arrangements, the pathway **180** may comprise any suitable number of lateral portions and offset axial portions. The additional lateral and offset axial portions may further improve the mixing of the supplied gases.

(92) The positioning of the lateral flow portions **180a**, **180c**, and the offset axial portion **180b** can beneficially extend the mixing length of the bore **130** downstream of the location L at which the supply channels enter the bore **130**. Extending the mixing length of the bore **130** can also extend the mixing time of the gases supplied to the bore **130**, which can improve uniformity of deposition and improve device yield. In particular, the embodiment of FIGS. 6A-6J can provide a mixing non-uniformity of less than 2%, e.g., less than 1%. In some embodiments, the extended mixing length provided by FIGS. 6A-6J can provide a mixing non-uniformity of less than 0.5%, or more particularly less than 0.15%, e.g., about 0.09%.

(93) For example, as shown in FIG. 6C, the manifold **100** can extend along a manifold length l. In some semiconductor devices, such as those illustrated in FIGS. 1-4F, the bore can extend generally along the axis Z from the inlet **120** to the outlet **132**. In such arrangements, therefore, the length of the bore can extend from a first end portion at the inlet **120** to a second end portion at the outlet **132** such that the length of the bore is the same as the length l of the manifold **100**. However, in the embodiment shown in FIGS. 6A-6J, the length of the gas pathway defined by the bore **130** from the inlet **120**, through the upstream axial portion **130A**, through the extended length mixing pathway **180**, and through the downstream axial portion **130B** may be greater than the length l defined between the inlet **120** and outlet **132** (e.g., between the first and second end portions of the

manifold **100**).

(94) FIG. 7A is a flowchart illustrating a method **700** for depositing one or more layers on a substrate. The method **700** begins in a block **701** to provide a manifold comprising a bore therein. The bore can define a gas passageway between a first end portion of the manifold and a second end portion of the manifold. The first end portion can be disposed opposite to and spaced from the second end portion along a longitudinal axis of the manifold by a first distance. In various embodiments, the first end portion of the manifold can comprise the end portion at which an inert gas inlet is disposed. The second end portion can comprise the end portion at which the outlet is disposed.

(95) Turning to a block **702**, a reactant gas can be supplied to the bore. In some embodiments, the reactant gas can be supplied to a distribution channel from a gas source. The gas can be conveyed to the bore by way of one or more supply channels extending from the distribution channel to the bore. In some embodiments, the supply channel can comprise a slit defining an at least partially annular gap through an inner wall of the bore. In other embodiments, the supply channels can comprise angled passageways that angle inwardly from the distribution channel to the bore at an acute angle.

(96) In a block **703**, the reactant gas can be directed along the gas passageway from the first end portion to the second end portion for a second distance. The second distance can be larger than the first distance. As explained herein, in some embodiments, the reactant gas can be directed along a first lateral portion extending non-parallel to and away from the longitudinal axis of the manifold. An offset axial portion of the pathway can convey the gas along the longitudinal axis. A second lateral portion can extend non-parallel to and towards the longitudinal axis of the manifold. In some arrangements, a downstream axial portion of the bore can convey the mixed gases to the reaction chamber. Advantageously, as explained herein, the extended mixing length can improve mixing and reduce non-uniformities of deposition processes.

(97) FIG. 7B is a flowchart illustrating a method **750** for depositing one or more layers on a substrate, according to various embodiments. In a block **751**, a manifold comprising a bore can be provided. The bore can comprise an axial portion that defines a longitudinal axis of the manifold and a lateral portion extending non-parallel to the longitudinal axis. In a block **752**, a reactant gas can be supplied to the axial portion of the bore at a first location along the longitudinal axis. For example, as explained above, the gas can be supplied to the axial portion by way of a supply channel comprising a slit. In other embodiments, the gas can be supplied to the axial portion by way of one or more angled supply channels.

(98) In a block **752**, the reactant gas can be directed through the axial portion of the bore parallel to the longitudinal axis. In a block **753**, downstream of the axial portion, the reactant gas can be directed through the lateral portion of the bore in a direction non-parallel to the longitudinal axis. In some embodiments, the gas can pass from the lateral portion into an offset axial portion of the bore in a direction parallel to (or including a directional component parallel to) the longitudinal axis. As explained herein, a second lateral portion can extend laterally towards the longitudinal axis to convey the gas from the offset axial portion to a downstream axial portion. The gas can be conveyed along the downstream axial portion to the reaction chamber.

IV. Manifolds with Single Gas Supply Tier

(99) Various embodiments disclosed herein relate to manifolds **100** with a single reactant supply channel for each reactant gas to be supplied to the bore **100**. For example, a semiconductor processing device can comprise a manifold comprising a bore defining an inner wall and a channel through the manifold. The device can include a source of a reactant gas. A supply channel can be configured to deliver the reactant gas to the bore by way of an opening on the inner wall of the bore. All the reactant gas can be delivered to the bore by the opening.

(100) FIG. 8A is a schematic perspective view of a manifold **100** that includes a single tier of gas delivery to the bore **130**. Unless otherwise noted, reference numerals in FIGS. 8A-8F represent

components that are the same as or similar to like numbered components in FIGS. 1-7B. For example, as shown in FIG. 8A, the manifold **100** can comprise a manifold body **102** connected with two blocks **112a**, **112b**. First and second valves **116a**, **116b** (e.g., reactant gas valves) may be disposed on the blocks **112a**, **112b**. The inert gas inlet **120** can supply inert gas to the bore **130**. (101) FIG. 8B is a schematic side cross-sectional view of a semiconductor processing device **10** comprising the manifold **10** of FIG. 8A and the reaction chamber **810**. FIG. 8C illustrates the flow pathway of the gases through the manifold **100** of FIG. 8B. As with the embodiment of FIGS. 6A-6J, the manifold **100** can comprise an upstream axial flow portion **130A**, an extended mixing length passageway **180**, and a downstream axial flow portion **130B**. In addition, the pathway **180** can comprise a first lateral portion **180a**, an offset axial portion **180b**, and a second lateral portion **180c**. Beneficially, the pathway **180** can extend the mixing length downstream of the location at which the gases enter the bore **130**. The extended mixing length can improve the mixing of supplied gases to improve uniformity and device yield.

(102) The extended mixing length provided by the pathway **180** can also advantageously enable the use of a single gas supply tier **190**. Unlike the embodiments of FIGS. 2-6J, in which the gases are supplied along multiple tiers spaced apart along the longitudinal axis Z, in the embodiment of FIGS. 8B-8C, the supply tier **190** can comprise a first supply channel **190a** and a second supply channel **190b** that delivers gases to the bore **130** by way of corresponding openings **191a**, **191b** in the inner wall of the manifold **100**. The first and second supply channels **190a**, **190b** can be disposed opposite one another at approximately the same axial location. In other arrangements, disposing the supply channels in a spaced relationship along the bore **130** can improve mixing. However, in FIGS. 8B-8C, the extended mixing length provided by the pathway **180** can enable use of the first supply channel **190a** to supply all the reactant gas provided to the manifold **100** by a corresponding first gas source. Similarly, the second supply channel **190b** can supply all the reactant gas provided to the manifold **100** by a second gas source. Thus, all the first gas can pass through the first supply channel **190a** and first opening **191a** to the bore **130**, e.g., to the upstream axial portion **130A**. All the second gas can pass through the second supply channel **190b** and second opening **191b** to the bore **130**, e.g., to the upstream axial portion **130A** of the bore **130**.

(103) FIGS. 8D-8F are schematic side cross-sectional views of the supply tier **190** shown in FIGS. 8B-8C, according to various embodiments. In FIG. 8D, the gas supply tier **190** comprises first and second supply channels **190a**, **190b** that extend non-parallel to the longitudinal axis Z, e.g., the supply channels **190a**, **190b** are disposed perpendicular to the axis Z. Gas flows through the openings **191a**, **191b** to the bore **130**, and are conveyed axially along the axis Z to the extended mixing pathway **180** and the reaction chamber **810**. In FIG. 8D, the supply channels **190a**, **190b** extend horizontally such that the channels **190a**, **190b** are perpendicular to the bore **130**. Such an arrangement can advantageously simplify manufacturing processes as compared with manifolds that utilize angled channels. The extended mixing length pathway **180** can facilitate downstream mixing of the supplied gases.

(104) FIG. 8E illustrates first and second supply channels **190a**, **190b** that have downwardly-angled portions **192** that convey the gases to the bore **130**. Angling the portions **192** along the axis Z can beneficially enhance mixing in the axial portion **130A** of the bore **130**. In FIG. 8F, the angled portions **192** enter the bore **130** upstream of an expanded bore portion **193** that has a larger diameter than the portion of the bore **130** upstream of the expanded portion **193**. The expanded portion **193** can comprise a mixing chamber in the bore **130** in which the supplied gases can mix together.

(105) FIG. 9 is a flowchart illustrating a method **900** of depositing one or more films on a substrate, according to various embodiments. In a block **901**, a manifold can be provided. The manifold can comprise a bore having an inner wall and defining a channel through the manifold. In a block **902**, all of a reactant gas can be supplied through the single opening on the inner wall of the bore. As explained herein, the use of a single gas supply tier can simplify manufacturing of the manifold.

For example, a second supply channel and a second opening can be disposed at the same axial position as the single opening and a first supply channel that supplies gas to the single opening. (106) Although the foregoing has been described in detail by way of illustrations and examples for purposes of clarity and understanding, it is apparent to those skilled in the art that certain changes and modifications may be practiced. Therefore, the description and examples should not be construed as limiting the scope of the invention to the specific embodiments and examples described herein, but rather to also cover all modification and alternatives coming within the true scope and spirit of the invention. Moreover, not all of the features, aspects and advantages described herein above are necessarily required to practice the present invention.

Claims

1. A semiconductor processing device, comprising: a manifold comprising a bore and having an inner wall, the inner wall at least partially defining the bore, a first axial portion of the bore extending along a longitudinal axis of the manifold; and a supply channel that provides fluid communication between a gas source and the bore, the supply channel comprising a slit defining an at least partially annular gap through the inner wall of the manifold to deliver a gas from the gas source to the bore, the at least partially annular gap revolved about the longitudinal axis, wherein the at least partially annular gap is revolved about the longitudinal axis in a range of 90° to 360°.
2. The semiconductor processing device of claim 1, further comprising a gas distribution channel that conveys the gas from the gas source to the supply channel, the gas distribution channel having a thickness along the longitudinal axis that is larger than the at least partially annular gap.
3. The semiconductor processing device of claim 2, further comprising a reactant gas valve configured to selectively transfer the gas to the gas distribution channel.
4. The semiconductor processing device of claim 2, wherein the thickness of the gas distribution channel is at least twice a thickness of the at least partially annular gap.
5. The semiconductor processing device of claim 1, wherein the at least partially annular gap is revolved about the longitudinal axis in a range of 120° to 360°.
6. The semiconductor processing device of claim 1, wherein a thickness of the at least partially annular gap is in a range of 0.1 mm to 1 mm.
7. The semiconductor processing device of claim 1, further comprising a second supply channel that provides fluid communication between a second gas source and the bore, the second supply channel comprising a second slit defining a second at least partially annular gap through the inner wall of the manifold to deliver a second gas from the second gas source to the bore, the second at least partially annular gap revolved about the longitudinal axis and disposed upstream of the at least partially annular gap.
8. A semiconductor processing device, comprising: a manifold comprising a bore; and a supply channel that provides fluid communication between a gas source and the bore to supply a gas to the bore, wherein the bore comprises a channel having an annular flow portion with an at least partially annular cross-section and a non-annular flow portion with a non-annular cross-section, the non-annular flow portion disposed downstream of the annular flow portion, wherein the manifold comprises a plurality of blocks connected together to define the bore and the supply channel, wherein the annular flow portion comprises a plug disposed in the bore, the annular flow portion disposed between an outer periphery of the plug and an inner wall of the manifold, and wherein the annular flow portion comprises a first block of the plurality of blocks, the first block comprising an opening through which the plug is disposed and one or more holes disposed adjacent the opening.
9. The semiconductor processing device of claim 8, wherein the supply channel comprises a slit defining an at least partially annular gap through the inner wall.
10. The semiconductor processing device of claim 8, wherein the plug comprises one or more tapered portions.

11. The semiconductor processing device of claim 8, wherein the bore comprises a second non-annular flow portion disposed upstream of the annular flow portion.
 12. The semiconductor processing device of claim 8, wherein the plurality of blocks connected together defines a slit formed by at least a partially annular gap through the inner wall of the manifold.
 13. The semiconductor processing device of claim 9, wherein the at least partially annular gap is revolved about a longitudinal axis in a range of 90° to 360.
 14. The semiconductor processing device of claim 8, further comprising a second supply channel that provides fluid communication between a second gas source and the bore, the second supply channel disposed upstream of the supply channel.
 15. A semiconductor processing device, comprising: a manifold comprising a bore; a supply channel that provides fluid communication between a gas source and the bore to supply a gas to the bore, the supply channel comprising a slit defining an at least partially annular gap through the inner wall of the manifold to deliver a gas from the gas source to the bore, the at least partially annular gap revolved about the longitudinal axis; and a plug disposed within the bore, wherein the bore comprises a channel having an annular flow portion with an at least partially annular cross-section and a non-annular flow portion with a non-annular cross-section, the non-annular flow portion disposed downstream of the annular flow portion.
 16. The semiconductor processing device of claim 15, wherein the slit supplies the gas to the annular flow portion of the bore.
 17. The semiconductor processing device of claim 15, wherein the annular flow portion is disposed between an outer periphery of the plug and the inner wall of the manifold.
 18. The semiconductor processing device of claim 15, further comprising a second supply channel that provides fluid communication between a second gas source and the bore, the second supply channel comprising a second slit defining a second at least partially annular gap through the inner wall of the manifold to deliver a second gas from the second gas source to the bore, the second at least partially annular gap revolved about the longitudinal axis and disposed upstream of the at least partially annular gap.
 19. The semiconductor processing device of claim 15, wherein the bore comprises a second non-annular flow portion disposed upstream of the annular flow portion.
 20. The semiconductor processing device of claim 15, further comprising a gas distribution channel that conveys the gas from the gas source to the supply channel.
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